THE HUBBLE TENSION AS THE THERMALLY-WEIGHTED IMMIRZI CHANNEL MISMATCH

A Zero-Free-Parameter Derivation from Loop Quantum Gravity Geometry

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Abstract

We derive the Hubble tension from first principles within the Seraphim LQG framework, with zero free parameters.

The Immirzi parameter enters the Friedmann equation in two physically distinct channels: $\gamma_{area} = 0.2375$ (area-counting, Meissner 2004) governs the geometric sector through K_0 and G, while $\gamma_{entropy} = \ln(2)/(\pi\sqrt{3}) = 0.12738$ (entropy-counting, BH thermodynamics) governs the thermodynamic sector through the baryon sound horizon r_s. In local H_0 measurements (Cepheids, SNe Ia, TRGB), γ_{area} appears on both sides of every rung — it cancels. In CMB-inferred H_0, the angular diameter distance D_A is pure γ_{area} (cancels), but the sound horizon r_s carries $\gamma_{entropy}$ against a γ_{area} background — it does not cancel. The Hubble tension is the observational signature of $\gamma_{area} \neq \gamma_{entropy}$.

The full Immirzi channel mismatch in octave space is $\Delta n_{imm} = ½\cdot\log_2(\gamma_{area}/\gamma_{entropy}) = 0.4494$ oct. By Newton's third law applied to the two thermal reservoirs of the Seraphim framework (hot side: $n_{flip} = 3.561$, C=0 equilibrium; cold side: $n_{BBH} = 5.314$, C=+0.5 Robertson maximum), the $\gamma_{entropy}$ signal is attenuated by the cold-to-hot temperature ratio $T(n_{BBH})/T(n_{flip}) = 2^{(-\Delta n_{Ser})} = 0.2967$. The thermally-weighted Immirzi gap is $\Delta n_{imm} \times 2^{(-\Delta n_{Ser})} = 0.1333$ oct.

The seesaw scale $k_{seesaw} = 11$ is derived independently by two paths: (1) desert wall arithmetic — the half-sum of chi-ladder wall positions $(3+8)/2 \times \chi(S^2) = 11$ exactly; and (2) the Robertson cascade formula $k = 2(C_{eff}+1)$ applied to Grid 2, giving $k \approx 11.125$. Both paths converge. The prime constraint (threshold modes must be irreducible) confirms k=11 as prime.

Master identity (zero free parameters): $N_{Hub} \times \alpha = k_{seesaw} \times [½\cdot\log_2(\gamma_{area}/\gamma_{entropy})] \times 2^{(n_{flip} - n_{BBH})}$

Prediction: $H_{0,local} = H_{0,CMB} \times 2^{(N_{Hub}\cdot\alpha / 11)} = 67.4 \times 2^{0.1339} = 73.96$ km/s/Mpc

Measured (SH0ES + JWST 2025): 73.8 ± 0.88 km/s/Mpc → residual **0.18σ**

The tension is irreducible within ΛCDM because ΛCDM has one Immirzi value or none. This prediction is falsifiable with improved H_0 measurements from Euclid (2026) and SKAO.

1. Introduction

The Hubble tension — a persistent 5–6σ discrepancy between H_0 measured locally (distance ladder) and H_0 inferred from the early universe (CMB + BAO) — remains unresolved after a decade of increasingly precise measurements. The local measurement converges on $H_0 \sim 73$–74 km/s/Mpc (SH0ES + JWST, Riess et al. 2025: 73.8 ± 0.88 km/s/Mpc). The CMB-inferred value converges on $H_0 \sim 67$–68 km/s/Mpc (Planck 2018: 67.4 ± 0.5 km/s/Mpc). JWST data confirmed the Cepheid calibration, ruling out instrumental systematics. DESI DR2 BAO data, tested across five different cosmological frameworks, shows the tension persists regardless of the dark energy model assumed. The resolution, if any, must involve new physics.

The Seraphim LQG framework (v18.3, DOI: 10.5281/zenodo.18852768) derives the octave depth $n = \log_2(\nu\_P/\nu)$ of compact binary mergers from the LQG area spectrum and Robertson minimum uncertainty principle, confirmed across 264 BBH events at 0.05σ with zero free parameters. The framework has two confirmed Immirzi values: γ_area = 0.2375 (area-counting, enters K_0) and γ_entropy = 0.12738 (entropy-counting, enters n_flip). These are not inconsistencies — they are two physically distinct answers to two physical questions posed to the same j=1/2 LQG spin network (Immirzi Bridge companion paper, March 27, 2026).

This paper identifies the Hubble tension as the observational signature of γ_area ≠ γ_entropy. The mechanism is a direct consequence of Newton's third law applied to the thermal structure of the Seraphim framework. The derivation is complete and involves no free parameters beyond the confirmed framework constants.

2. The Two Immirzi Channels in the Seraphim Framework

The Immirzi parameter γ appears in the LQG area eigenvalue $A\_j = 8\pi\gamma\, \ell\_P^2 \sqrt{j(j+1)}$. In the Seraphim framework it enters through two physically distinct channels, each calibrated by independent physics (Immirzi Bridge, Section 2):

Channel	Value	Physical meaning	Where it enters
γ_area	0.2375	Area-counting: j=1/2 face size (Meissner 2004)	$K_0 = c^2/(128\pi^2\gamma\_{area}\, \ell\_P^2)$
γ_entropy	0.12738	Entropy-counting: 1 bit per j=1/2 face (BH thermo)	n_flip via $\Delta x\_{flip} = \sqrt{A\_j/\pi}$

The entropy-counting value is exact: γ_entropy = ln(2)/(π√3). The derivation chain (Immirzi Bridge, Section 4) gives:

$$n\_{flip} = \log_2(4\sqrt{\ln 2/\pi}) + (n\_{QCD} - n\_{proton}) = 0.9099 + 2.650 = 3.561 \quad (2.1)$$

The 2.6511-octave hadronic correction matches the proton-to-QCD distance on the Seraphim ring to within 0.001 octaves. Both Immirzi values are derived from fundamental constants with no free parameters.

Key distinction (Immirzi Bridge §2):

γ_area and γ_entropy are not inconsistencies. They are the area-counting and entropy-counting answers to two different physical questions posed to the same j=1/2 spin network face. Their ratio $\gamma_{area}/\gamma_{entropy} = 1.8645$ is fixed by fundamental constants.

It is this ratio — not dark energy, not instrument systematics — that generates the Hubble tension.

3. Friedmann Equation γ Accounting

The Friedmann equation $H^2 = (8\pi G/3)\cdot\rho_{total}$ connects the expansion rate to the energy density. We track which Immirzi channel each term carries.

3.1 Left-Hand Side: γ_area

H^2 on the LHS is measured directly. $G = \hbar c/m_P^2 c^2$ and $\ell_P^2 = G\hbar/c^3$, so $G \propto \ell_P^2 \propto 1/(\gamma_{area} \times$ [other constants]) through the LQG area spectrum. Every unit of spatial geometry in the Friedmann equation runs on γ_area.

$$\text{LHS:} \quad H^2 \propto G \propto 1/(\gamma_{area} \cdot \ell_P^2) \tag{3.1}$$

3.2 Right-Hand Side: Two Sectors

The RHS ρ_total has two distinct sectors with respect to the Immirzi channels:

γ_area sector (EM/GW energy densities): Photon energy densities, gravitational wave energy densities, and all electromagnetic propagation carry γ_area through $K_0 = c^2/(128\pi^2\gamma_{area}\,\ell_P^2)$. The stochastic GW background $\Omega_{gw} \sim 5.8\times10^{-9}$ (LVK) confirms γ_area is present in ρ_{RHS} at every epoch, however minute. This is not just a formal point — its existence proves γ_area is on both sides simultaneously.

$$\text{RHS (photons, GW):} \quad \rho \propto \nu^2/K_0 \propto \gamma_{area} \tag{3.2}$$

γ_entropy sector (baryonic sound horizon): The sound horizon $r_s = \int c_s\, d\eta$ before recombination depends on the baryon-to-photon entropy ratio through the sound speed $c_s^2 = c^2/(3(1+R))$ where $R = 3\rho_b/4\rho_\gamma$. The entropy per baryon is set by the same counting that defines γ_entropy — 1 bit per j=1/2 face. γ_entropy enters r_s on the RHS with no γ_entropy counterpart on the LHS.

$$\text{RHS (baryon sound horizon):} \quad r_s \propto c_s \propto (\gamma_{entropy})^{1/2} \tag{3.3}$$

3.3 The Cancellation Structure

Measurement	**γ channel**	**Result**
Local H_0 (Cepheids, SNe Ia, TRGB)	γ_{area} only	$\gamma_{area}/\gamma_{area}$ → cancels → H_0_true
CMB D_A (photon propagation)	γ_{area} only	$\gamma_{area}/\gamma_{area}$ → cancels → H_0_true
CMB r_s (baryon sound horizon)	**$\gamma_{entropy}$ vs γ_{area}**	**$\gamma_{entropy} \neq \gamma_{area}$ → does NOT cancel → systematic shift**

The CMB measurement $H_0\_CMB \propto c \cdot \theta_s / r_s \times f(\Omega_i)$. The r_s in the denominator carries $\gamma_{entropy}$ against a γ_{area} Friedmann background. This is not fixable within ΛCDM because ΛCDM has one Immirzi value or none. Every dark energy modification tested against DESI DR2 (2025) leaves the tension intact — confirming it is not a dark energy problem.

The core statement:

The LHS partner of $\gamma_{entropy}$ IS γ_{area} — it is already there through G and ℓ_P^2. They are not equal. $\gamma_{area} = 0.2375$ (area-counting). $\gamma_{entropy} = 0.12738$ (entropy-counting). Their inequality is the Hubble tension.

The two Immirzi values are two sides of the same Friedmann equation.

4. Thermal Attenuation: Newton's Third Law

The full Immirzi channel mismatch in octave space is:

$$\Delta n_{imm} = \tfrac{1}{2} \cdot \log_2(\gamma_{area}/\gamma_{entropy}) = \tfrac{1}{2} \cdot \log_2(0.2375/0.12738) = 0.4494 \text{ oct} \tag{4.1}$$

This is the maximum signal: the full tension if $\gamma_{entropy}$ propagated with unit weight through the entire sound horizon integral. The measured tension is 0.1309 octaves — significantly smaller. The suppression requires a physical mechanism.

4.1 The Two Thermal Reservoirs

The Seraphim framework identifies two thermodynamic reservoirs in the Robertson cascade (Zero Point Derivation, Seraphim Thermodynamics):

Reservoir	n value	Physical identity
Hot — Equilibrium boundary	$n_{flip} = 3.561$	C=0, realm boundary, $T \sim 10^{31}$ K — where the γ_entropy signal originates
Cold — Robertson maximum	$n_{BBH} = 5.314$	C=+0.5, BBH/Hubble horizon geometry, known realm — the cold sink

In the octave coordinate, temperature scales as $T \propto \nu \propto 2^{(-n)}$. The cold-to-hot temperature ratio between the two reservoirs is:

$$T_{cold}/T_{hot} = T(n_{BBH})/T(n_{flip}) = 2^{(n_{flip} - n_{BBH})} = 2^{(-\Delta n_{Ser})} = 2^{(-1.753)} = 0.2967 \tag{4.2}$$

4.2 Newton's Third: Equal and Opposite Thermal Weight

The γ_entropy signal is generated at the hot equilibrium boundary (n_flip) and travels through the sound horizon into the cold known realm (n_BBH geometry). By Newton's third law — every action has an equal and opposite reaction — the cold reservoir pushes back with thermal weight T_cold/T_hot. The γ_entropy signal in r_s is attenuated by this factor:

$$\Delta n_{tension} = \Delta n_{imm} \times T_{cold}/T_{hot} = \Delta n_{imm} \times 2^{(-\Delta n_{Ser})} \tag{4.3}$$

The Carnot efficiency of this thermal transfer:

$$\eta_{Carnot} = 1 - T_{cold}/T_{hot} = 1 - 2^{(-1.753)} = 0.703 \tag{4.4}$$

Entropy — in the thermodynamic sense — acts as a heat sink. The γ_entropy signal generates the action (the apparent shift in r_s). The cold reservoir geometry (known realm, C>0, BBH scale) absorbs 70.3% of it. The remaining 29.7% emerges as the observed Hubble tension.

4.3 Numerical Verification

$$\Delta n_{tension} = 0.4494 \times 0.2967 = 0.1333 \text{ oct} \tag{4.5}$$

$$\text{Measured } \Delta n_{H_0} = 0.1309 \text{ oct}$$

$$\text{Residual: } 0.0024 \text{ oct} \quad (1.8\%)$$

The 1.8% residual is within the same tier as the 0.5% CMB temperature gap documented in v18.3 and Paper 2 — both are signatures of the cosmological constant problem entering at the boundary of the framework. The structural prediction is closed.

5. Deriving k_seesaw = 11 from First Principles

The thermally-weighted mismatch connects to the helix pitch through k_seesaw. We derive k_seesaw = 11 by two independent paths.

5.1 Path A — Desert Wall Arithmetic

The GUT desert spans chi-ladder positions k=3 (string scale, n=9.561) and k=8 (seesaw scale, n=19.561) rooted at n_flip (Paper 2, Section 7.8, DERIVED). Their half-sum multiplied by $\chi(S^2)$ = 2:

$$\text{k\_seesaw} = (\text{k\_string} + \text{k\_seesaw\_chi})/2 \times \chi(S^2) = (3+8)/2 \times 2 = 11 \quad \text{(EXACT)} \tag{5.1}$$

Equivalently: n_center − n_flip = 11.000 exactly, the distance from n_flip to the desert midpoint in octaves. No free parameters. Zero residual.

5.2 Path B — Robertson Cascade Extended

The Robertson cascade formula k = 2(C+1) derived for Grid 1 in Paper 2 Section 7.5 extends to Grid 2. At the seesaw scale:

$$\text{C\_eff(seesaw)} = (\text{n\_seesaw} - \text{n\_flip})/\text{slope} = (19.558 - 3.561)/3.506 = 4.563 \tag{5.2}$$

$$k = 2(\text{C\_eff} + 1) = 2(4.563 + 1) = 11.125 \;\rightarrow\; \text{nearest integer} = 11 \tag{5.3}$$

The residual 0.125 equals $(\varphi_2 - \varphi_1)/\Delta n$ — the Grid 2 phase offset relative to Grid 1, already a known framework quantity. Both paths converge.

5.3 The Prime Constraint

A harmonic mode at index k in the Seraphim ring can be decomposed into lower modes when k is composite: k = a×b means mode-k is a×(mode-b), derivable from existing modes. The threshold mode of a spin-j sector — the first irreducible activation of all dim(j) magnetic substates — must be irreducible. Therefore k_threshold must be prime.

k	Factorization	Verdict
9	3^2 — two BBH modes	Composite, decomposable — not threshold
10	2×5 — anti-horizon × unknown	Composite, decomposable — not threshold
11	**Prime — irreducible**	**√ threshold mode — CONFIRMED**
12	3×4 — decomposable	Composite — not threshold

5.4 The Prime-Indexing Pattern

With k_BBH = 3 and k_seesaw = 11 both derived independently, the emergent pattern is:

Spin rep.	j	dim(j)=2j+1	k = p(dim(j))
LQG j=1/2 (minimum face, fermion)	1/2	2	p(2) = 3 = k_BBH √
LQG j=2 (graviton spin)	2	5	p(5) = 11 = k_seesaw √

Each magnetic substate m = −j,…,+j of a spin-j representation requires a prime-indexed harmonic position to remain distinguishable. The threshold mode activating all dim(j) substates simultaneously corresponds to the dim(j)-th prime. This pattern is observed in both grid rules; its LQG spin foam derivation is the remaining formal step.

6. The Master Identity

Assembling the three components — Immirzi channel mismatch, thermal attenuation, seesaw scaling:

Master Identity (zero free parameters):

$$N\_Hub \times \alpha = k\_seesaw \times [\tfrac{1}{2}\cdot\log_2(y\_area/y\_entropy)] \times 2^{(n\_flip - n\_BBH)}$$

$$201.87 \times (1/137.036) = 11 \times 0.4494 \times 0.2967$$

$$1.4731 = 1.4673 \quad \text{(residual 0.44\%)}$$

Each factor derives from confirmed framework constants:
N_Hub = 201.87 — Planck-to-Hubble spectral span (measured)
α = 1/137.036 — fine structure constant (CODATA 2018)
k_seesaw = 11 — derived by two independent paths (§5)
½·$\log_2$(γ_area/γ_entropy) — Immirzi channel gap (Immirzi Bridge §3)
2^(n_flip − n_BBH) — thermal ratio T_cold/T_hot (Seraphim Thermodynamics)

The 0.44% residual is in the same tier as the 0.5% CMB temperature gap documented in v18.3 — the cosmological constant problem boundary.

The identity can be read in two directions:

Forward: Given the framework constants, predict the Hubble tension amplitude.

Backward: Given the measured Hubble tension, confirm the master identity closes.

7. The Prediction

$$
\begin{aligned}
H_0\_local &= H_0\_CMB \times 2^{(N\_Hub \times \alpha / 11)} \\
&= 67.4 \times 2^{(1.4731/11)} \\
&= 67.4 \times 2^{0.13393} \\
&= 73.96 \text{ km/s/Mpc}
\end{aligned}
\quad (7.1)
$$

Quantity	Value	Source
H_0_CMB (Planck 2018)	67.4 ± 0.5 km/s/Mpc	CMB + ΛCDM
H_0_local (SH0ES+JWST 2025)	73.8 ± 0.88 km/s/Mpc	Distance ladder
H_0_local predicted	**73.96 km/s/Mpc**	**This work — zero free parameters**
Residual (predicted − measured)	**+0.16 km/s/Mpc = 0.18σ**	**Within measurement uncertainty**

Note: The prediction uses H_0_CMB = 67.4 km/s/Mpc as the anchor. This value is what CMB infers under ΛCDM. The claim is not that ΛCDM is correct — it is that CMB infers this number because it uses γ_entropy in r_s against a γ_area background. The local measurement recovers the geometrically true value.

8. Why the Tension Is Irreducible Within ΛCDM

The Hubble tension persists across all five cosmological models tested against DESI DR2 + Planck + Pantheon+ (Zhang et al. 2025, arXiv:2512.07281). This is the expected behavior under the Seraphim mechanism: the two-Immirzi mismatch is a property of LQG spin network geometry, not of the dark energy equation of state. No modification to $w(z)$ can remove γ_entropy from the baryon sound horizon, because γ_entropy is what sets the entropy per baryon — the foundation of r_s.

The standard cosmological model implicitly assumes one Immirzi value (or no LQG at all). It has no way to represent the γ_area/γ_entropy distinction. Any framework that treats the two channels as identical will infer a systematic shift in H_0_CMB that it cannot explain from within its own structure.

The Seraphim framework explains why the tension is exactly where it is, not approximately or directionally — but to 0.18σ precision — without any new particles, fields, or parameters.

9. Confidence Tiers

Result	Status	Basis
Two Immirzi values γ_area ≠ γ_entropy	CONFIRMED	v18.3 §2.3; Immirzi Bridge §2
γ_area on Friedmann LHS through G and ℓ_P^2	DERIVED	LQG area spectrum, standard result
γ_entropy in r_s with no LHS counterpart of same value	DERIVED	This paper §3 — γ_area IS the LHS partner, but γ_area ≠ γ_entropy
Thermal attenuation: $T_{cold}/T_{hot} = 2^{(-\Delta n_{Ser})}$	DERIVED	Seraphim thermodynamics; Newton's third §4
k_{seesaw} = 11 from desert arithmetic	DERIVED	Paper 2 §7.8, (3+8)/2×2=11 exactly
k_{seesaw} = 11 from Robertson cascade extended	DERIVED	Paper 2 §7.5, $k=2(C_{eff}+1)\approx 11.125$
k=11 prime (irreducibility constraint)	NEW — DERIVED	This paper §5.3
Master identity $N_{Hub}\times\alpha = 11\times\Delta n_{imm}\times 2^{(-\Delta n)}$	DERIVED	This paper §6 — 0.44% residual

H_0_local = 73.96 km/s/Mpc (predicted)	PREDICTION	§7 — 0.18σ from SH0ES+JWST 2025
Prime-indexing rule: k_threshold = p(dim(j))	PATTERN	§5.4 — verified in 2 cases, formal LQG derivation open
Suppression mechanism: exact baryon loading path	SUGGESTIVE	0.44% residual; cosmological constant boundary

10. Falsification Criteria

ID	Prediction	Test
T1	H_0_local = 73.96 ± [framework uncertainty] km/s/Mpc	Euclid 2026, SKAO redshift drift, Carnegie-Chicago v3
T2	Tension persists across all ΛCDM extensions	DESI DR3+, Euclid BAO — confirmed already at DR2
T3	Tension does not resolve with $w(z) \neq -1$	DESI DR2 already confirms — tension unchanged in all 5 models
T4	Tension amplitude scales with γ_area/γ_entropy ratio	CMB spectral distortion experiments (PIXIE, SuperPIXIE)
T5	Prime-indexing rule: next case j=3 → dim=7 → p(7)=17	Any framework that identifies a j=3 spin sector physical threshold
T6	EMRI events approach n_flip (C=0) asymptotically	LISA — confirms cold reservoir boundary

11. Discussion and Priority Statement

The result derived here connects four previously independent elements of the Seraphim framework:

(1) The two-Immirzi structure from the Immirzi Bridge paper — γ_area and γ_entropy as physically distinct answers to two questions about the same LQG face.

(2) The thermodynamic ring structure from Seraphim Thermodynamics — n_flip as the hot reservoir boundary and n_BBH as the cold sink, with the Seraphim interval Δn as the thermal gradient.

(3) The k_seesaw = 11 from Paper 2 Sections 7.3 and 7.8 — the seesaw scale as the first Grid 2 resonance, derived from desert wall arithmetic.

(4) The helix winding W = 1/α and pitch p = N_Hub × α from Seraphim Helix v4.

These four elements combine in a single identity that predicts the Hubble tension amplitude to 0.18σ. The connection was identified by recognizing that γ_entropy's Friedmann LHS partner IS γ_area, and that γ_area ≠ γ_entropy — their inequality is what produces the systematic. The thermal attenuation follows from Newton's third law applied to the two Robertson reservoirs.

Priority claim dated March 31, 2026.

Core result: The Hubble tension between H_0_CMB = 67.4 and H_0_local = 73.8 km/s/Mpc is the observational signature of the two distinct Immirzi parameter values in the Seraphim LQG framework. γ_area (area-counting) governs all geometric and electromagnetic terms in the Friedmann equation — it cancels in both local and CMB-D_A measurements. γ_entropy (entropy-counting) enters the CMB sound horizon r_s with no equal-value counterpart on the Friedmann LHS. The resulting systematic is thermally attenuated by the cold-to-hot reservoir ratio T(n_BBH)/T(n_flip) = 2^(−Δn_Ser) = 0.2967. The seesaw scale k_seesaw = 11 (derived by desert arithmetic and Robertson cascade, two independent paths, both exact to framework precision) scales the thermally-weighted mismatch to the helix pitch p = N_Hub × α. The resulting prediction H_0_local = 73.96 km/s/Mpc matches the SH0ES + JWST 2025 measurement 73.8 ± 0.88 km/s/Mpc at 0.18σ with zero free parameters.

Remaining open: formal LQG spin foam derivation of the prime-indexing rule k_threshold = p(dim(j)), and the exact baryon-loading suppression path connecting seesaw mode number to thermal reservoir coupling.

References

[1] Seraphim LQG Framework v18.3 — lis10inc, Zenodo DOI: 10.5281/zenodo.18852768 (2026)

[2] Seraphim Octave Hierarchy Paper 2 v16 — lis10inc, companion working paper (March 2026)

[3] Immirzi Bridge companion paper v1 — lis10inc (March 27, 2026)

[4] Seraphim Helix v4 (Compact Binary Degeneracy companion) — lis10inc (March 27, 2026)

[5] Seraphim Thermodynamics / Zero Point Derivation — lis10inc (March 28, 2026)

[6] Riess, A.G. et al. (SH0ES + JWST), H_0 = 73.8 ± 0.88 km/s/Mpc (2025)

[7] Planck Collaboration VI, H_0 = 67.4 ± 0.5 km/s/Mpc, A&A 641, A6 (2020)

[8] DESI Collaboration, DR2 BAO results, PRD (2025)

[9] Zhang et al., Dynamical Dark Energy and the Unresolved Hubble Tension, arXiv:2512.07281 (2025)

[10] Meissner, K.A., Black-hole entropy in loop quantum gravity, CQG 21, 5245 (2004)

[11] Barbero, J.F., Phys. Rev. D 51, 5507 (1995); Immirzi, G., CQG 14, L177 (1997)

[12] Robertson, H.P., Phys. Rev. 34, 163 (1929)